

CUSTOMER RETENTION ANALYTICS – EXECUTIVE SUMMARY

Objective

The objective of this project is to analyze customer purchasing behavior, identify valuable customer segments, evaluate promotional dependency patterns, and develop actionable retention strategies using Python, SQL, and Power BI.

Dataset and Methodology

The dataset was cleaned and transformed using Python. Several engineered features were created, including Loyalty Band, Value Tier, Promo Dependency Level, Frequency Score, and Satisfaction Flag. SQL queries were used to answer key business questions, while Power BI was used to develop an interactive dashboard for business stakeholders.

Key Findings

- The dataset contains approximately **3,900 customers**.
- Total revenue generated is approximately **233K USD**.
- **VIP Loyalists** contribute the highest revenue among all customer segments.
- Customers are distributed across four value tiers: Bronze, Silver, Gold, and Platinum.
- A significant portion of customers exhibit **high promotional dependency**, indicating sensitivity to discounts.
- Loyalty-based segmentation reveals clear differences in customer engagement and purchasing behavior.

Business Recommendations

1. Gradually reduce discount dependency among VIP customers and focus on loyalty rewards.
2. Design re-engagement campaigns for Low Engagement customers.
3. Use targeted promotions instead of blanket discounts to improve marketing efficiency.
4. Prioritize Growth Customers for retention campaigns, as they have strong potential to become high-value customers.
5. Leverage customer segmentation for personalized marketing and improved customer lifetime value.

Conclusion

The project demonstrates how data analytics can be used to understand customer behavior, improve retention strategies, and support data-driven decision-making. The combination of Python, SQL, and Power BI provides a complete analytics workflow capable of generating meaningful business insights.